

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 40

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • High mean wind velocities that are consistent in strength and direction (1) • A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK (1)	2
01.2	<i>1 mark for each point.</i> • Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber; the air forced in and out drives a turbine (1) • Or point absorber (floating buoy moving relative to a fixed base turns a generator); surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine); overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines); oscillating wave surge converter (Oyster) (1)	2
01.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Equipment must withstand storms and corrosion (1) • Difficult to anchor off stormy coasts / in deep water (1) • Expensive to transmit electricity from isolated coasts to consumers (1) • Competition with established technologies (1)	2

Question	Marking guidance	Marks
01.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Barrage: huge quantities of construction materials; reduces tidal range so intertidal mudflats used by wading birds are lost; changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity; pollutants concentrate in the lagoon; blocks migratory fish and shipping (up to 2) • Lagoon: similar but smaller scale; only part of the estuary enclosed so the most sensitive areas and main channel can be avoided (up to 2) • In-stream turbines: do not block the tidal flow so little impact on range or sediment; but noise affects cetaceans and many dispersed turbines could have a larger total impact; harness less energy (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
02.1	<i>1 mark.</i> • The dose that kills 50% of the (test) population (1)	1
02.2	<i>1 mark for each point.</i> • Insecticide X: LD50 for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well (1) • Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) (1) • Y has a much higher LD50 for the beetle (150) so few predators killed / more specific (1)	3
02.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • High water solubility so it is leached into rivers / groundwater / runoff (1) • Toxic to aquatic organisms downstream (1) • Mobile so it can affect areas away from the field (1) • Washed off the crop quickly so more frequent application needed (1)	2
02.4	<i>1 mark for each point.</i> • Long half-life = high persistence so it is not broken down quickly (1) • So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble) (1)	2
02.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Absorbed and translocated within the plant so protects new growth (1) • Not washed off by rain (1) • Does not need to cover all leaf surfaces / less spray needed (1) • Less spray drift so less harm to non-target species (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	
03.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(4.8 - 1.4) / 4.8 \times 100$ (1) • = 70.8% (accept 71%) (1)	2
03.2	<i>1 mark for each point.</i> • A tributary joins the river, more than doubling the flow (15 to $36 \text{ m}^3 \text{ s}^{-1}$) (1) • So the pollutant is diluted by the extra water (dispersal), not broken down (1)	2

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> - The pollutant is persistent so it is not degraded / broken down as it travels (1) - Little extra water joins the river (flow 36 to 38) so little further dilution (1)	2
03.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away (1) - Smaller volume of water / less dilution / less dispersal by currents (1) - Persistent pollutant gradually becomes more concentrated (eg by evaporation) (1)	2
03.5	<i>1 mark for each point.</i> - Marine discharge: large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) (1) - River: small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	
04.1	<i>1 mark for each point.</i> - The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) (1) - Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) (1) - Lighter designs using less material; longer vehicle life spreading the embodied energy (1)	3
04.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Use of recyclable materials wherever possible (1) - Easy identification of components and their composition (code stamps) (1) - Easy dismantling and separation of components / materials (1) - Reusable components for new vehicles (1) - Compostable materials for parts that cannot be recycled (1)	4
04.3	<i>1 mark for each point.</i> - At low speed a large proportion of the energy just runs the engine rather than moving the car (1) - At high speed energy is wasted overcoming air resistance (which rises with the square of speed); optimum for many cars about 56 mph (1)	2
04.4	<i>1 mark.</i> Automatic stop / start systems cutting the engine when stationary; smooth driving in higher gears avoiding sudden braking / acceleration (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	

Question	Marking guidance	Marks
05.1	<i>1 mark for each point.</i> • Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction (1) • Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast (1) • The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur (1) • These support a rich food web including the anchovy fishery; the water warms as it travels west (1)	4
05.2	<i>1 mark for each point.</i> • The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east (1) • Upwelling of nutrient-rich cold water off South America stops (1) • Phytoplankton productivity collapses so the food web / anchovy population collapses; catch fell from about 7.7 to 1.7 million tonnes (1)	3
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Heavy rain and floods in the normally dry coastal deserts of Peru / South America (1) • Drought and reduced rainfall in eastern Australia / Indonesia (1) • Droughts in NE Africa, southern Africa, China (1) • Fewer hurricanes in the North Atlantic / fewer cyclones in Japan (1)	2
05.4	<i>1 mark.</i> • Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	
06.1	<i>1 mark for each point.</i> • Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve (1) • Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed (1) • Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle (1)	3
06.2	<i>1 mark for each point.</i> • Auger collects a core of known cross-sectional area and depth (1) • So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) (1)	2
06.3	<i>1 mark for each point.</i> • Minerals such as calcium carbonate break down releasing carbon dioxide (1) • So mass is lost that is not organic matter (1)	2

Question	Marking guidance	Marks
06.4	<i>1 mark for each point.</i> • A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter (1) • Standardise bulb brightness / colour, distance and container (1) • A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
07.1	<i>1 mark for each point.</i> • Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury (1) • Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes (1) • It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water (1) • Local people ate large amounts of fish (critical group) so received toxic doses (1)	4
07.2	<i>1 mark for each point.</i> • Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) (1) • It crosses the placenta and harms the developing unborn baby (1) • Mercury is a teratogen, preventing normal gene expression / development (1)	3
07.3	<i>1 mark for each point.</i> • Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs (1) • Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut (1) • Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
08.1	<i>Correct answer only.</i> • Oxygen (1)	1
08.2	<i>1 mark for each point.</i> • A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) (1) • There is still a lot that is not understood about how atmospheric processes work (1)	2
08.3	<i>1 mark for each point.</i> • The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes (1) • Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
09.1	<i>1 mark for each point.</i> • Solar PV cost fell dramatically from about 480 to 50 \$/MWh (about 90%), the steepest fall (1) • Wind fell steadily from 200 to 45 (1) • Gas stayed roughly constant at 60-70; solar and wind became cheaper than gas around 2015-2020 and are projected to fall further (1)	3
09.2	<i>1 mark for each point.</i> • Development / research costs are still being paid for and there are no economies of scale in manufacture (1) • They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure (1) • As output grows, mass production, improved designs / efficiency and experience reduce costs (1)	3
09.3	<i>1 mark for each point.</i> • Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid; running costs are very low (1) • Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments (1)	2
09.4	<i>1 mark for each point.</i> • Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user (1) • Some impacts (climate change) cannot yet be costed; decommissioning and waste costs are also uncertain (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter • Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity • Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NO_x, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	

Question	Marking guidance	Marks
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NO _x , biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NO_x).	
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
10.2	<i>1 mark.</i> • The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Question	Marking guidance	Marks
11.1	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25